

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	AI-Powered Sustainable Food Shopping Mobile Application
<i>Company:</i>	IBM
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐
	If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Mobile application development; AI-assisted recommendations; human-computer interaction; sustainability technology; data-driven decision support; food informatics; responsible AI; accessibility and inclusive design.
Project Background Information	
This project aims to develop a mobile application that assists users in making better and more sustainable choices when shopping for food. Using IBM Bob as part of the development process and, where appropriate, as part of the application experience, the app will provide personalised recommendations based on user preferences, dietary needs, and sustainability goals. The project addresses a real-world challenge: consumers often want to make lower-impact food choices but do not always have clear, accessible information at the point of purchase. The proposed app will help users understand product options, compare sustainability signals, and receive practical guidance while shopping. Students should design the application so that it is easy to use, informative, and engaging. The solution should support barcode scanning, real-time feedback, educational content, and recommendation features that help users make more informed choices. The project should also consider how AI can explain recommendations clearly and responsibly, avoiding unsupported claims and making any data limitations visible to users. Students must use IBM Bob during the development of the project. IBM Bob should support the software engineering workflow, for example by helping the team interpret requirements, create user stories, design the solution architecture, generate or review prototype code, develop test cases,	

¹ An overview of different research areas in the school of computer science can be found [here](#).

improve accessibility, and produce user documentation. The final report should explain how IBM Bob was used and provide evidence of meaningful use during development.

Requirements

The group must provide:

- A working mobile application prototype that helps users make better and more sustainable food shopping choices.
- Personalised recommendations based on user preferences, dietary restrictions, shopping behaviour, and sustainability goals.
- A barcode scanning feature that provides instant product information where suitable open product data is available.
- A real-time feedback feature that gives users immediate insight into the potential environmental impact or sustainability considerations of selected food items.
- Educational content such as short explainers, tips, and resources on sustainable food choices and practices.
- A clear approach for using IBM Bob in the development of the application and, where appropriate, in the functionality of the application.
- A recommendation and classification approach that is transparent, explainable, and bounded by the data available to the application.
- A database or local storage approach for user preferences, scanned items, and recommendation history, using privacy-by-design principles.
- API integration with suitable open food data sources, such as Food Repo, Open Food Facts, and/or USDA FoodData Central, subject to availability and appropriate use terms.
- A final technical report including architecture, data sources, recommendation logic, testing, privacy considerations, accessibility considerations, installation instructions, limitations, and recommendations for future development.
- A short demonstration video showing the app in use, including barcode scanning, recommendations, real-time feedback, educational content, and any IBM Bob-supported functionality.
- A GitHub repository or equivalent source-code submission containing the application, setup instructions, test evidence, and project documentation.

Objectives:

1. Develop a user-friendly mobile application that uses IBM Bob during development to support the creation of AI-assisted recommendation and explanation features.
2. Use AI-supported reasoning, classification, and explanation techniques to provide relevant information about food choices and sustainability considerations.
3. Enhance the user experience through interactive features such as barcode scanning, real-time feedback, and educational content on sustainable eating.

Technical Requirements:

- IBM Bob: Students must use IBM Bob during the development of the project and should document how it supported the development process.
- Mobile Development Framework: Choose a suitable framework, such as React Native or Flutter, for cross-platform development.
- Database Management: Implement appropriate storage for user preferences, scanned items, and recommendation history.
- API Integration: Integrate APIs for barcode scanning and access to sustainability, product, or nutrition data where available.
- Responsible AI: The app should clearly explain recommendations, handle uncertainty, and avoid presenting unsupported claims as fact.

Expected Outcomes:

- Functional Mobile Application: A working app that helps users make informed and sustainable food choices.
- User Engagement: Interactive and educational features that encourage repeated use and learning.
- Positive Environmental Impact: Support for more sustainable eating habits through clearer information and actionable recommendations.

• Development Evidence: Clear evidence that students used IBM Bob in requirements analysis, design, implementation, testing, and documentation.

Suggested Resources:

1. Food Repo API: open data on barcoded food products.
2. Open Food Facts API: open data on food products from around the world.
3. USDA FoodData Central API: nutrient data that can support product information and dietary features.

SkillsBuild Requirement:

It is expected that students will use IBM Bob in the development of the application and in its functionality where appropriate. Students should also complete three relevant IBM SkillsBuild AI badges and reference these in the final report.

Skills

Highly desirable:

- Mobile app development using React Native, Flutter, Swift, Kotlin, or similar.
- API integration and experience working with JSON/open data services.
- Database or local storage design for preferences, scanned products, and recommendation history.
- Practical use of IBM Bob as an AI development assistant.
- Interest in sustainability, food systems, nutrition, or responsible technology.

Desirable:

- Experience with recommendation systems, classification, or explainable AI.
- UI/UX design for consumer mobile applications.
- Barcode scanning, camera integration, or product lookup experience.
- Accessibility-aware design and inclusive user experience.
- Understanding of privacy-by-design, responsible AI, and data limitations.

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.